

9

Risk Management of Energy Derivatives

9.1 Introduction

Although energy price risk management is in the early stages of development compared to the financial markets, there is an increasing move of energy companies towards regular buying and selling of forwards, swaps, caps, and other option contracts. In this chapter we discuss the day-to-day management of price risk for institutions that trade energy derivatives and who are then faced with the problem of managing the risk of those positions. In chapters 10 and 11 we look at the risk management issues of calculating Value at Risk and measuring credit risk for energy derivative portfolios.

We can think of risk management as the immunisation of risk – this is achieved, for example, by setting up portfolios that contain positions in the underlying energies and energy derivatives in such a way that the portfolio is not affected by small changes in the price of the underlying energy and other key variables. We will see that the sensitivities of components of the portfolio to changes in their valuation parameters provide the key information for risk management.

For an energy company that sells an option position the most complete hedge is an exactly offsetting position, e.g., long calls bought against short calls sold. This long position is then held as a static hedge – meaning that the hedge need not be changed through time. However, in most circumstances this hedge may not be practical or profitable. Although it may be impossible to trade the offsetting position in this way, option positions can be created synthetically to give the same payoff. For example, as a first step the hedger makes their derivative portfolio immune to small changes in the underlying energy price in the next small interval of time. This process is known as delta hedging and the position is said to be delta-neutral. With a dynamic hedge the position will be neutral over a short horizon and the hedge must be rebalanced or rolled through time. Often option positions are also gamma and vega hedged, making the portfolio relatively insensitive to large changes in the underlying energy price and to changes in the volatility of the energy respectively. We also look in this chapter how hedging can be accomplished within the full multi-factor model of the previous chapter.

9.2 Delta Hedging of Energy Derivative Positions

The process of delta hedging an option involves dynamically trading a position in the underlying energy contract in such a way that, over each small interval of time between trades, the change in the option price is offset by an equal and opposite change in the value of the position in the underlying. In this way the hedged portfolio – consisting of the option and the position in the underlying – does not change in value, and is therefore immune to the risk of the underlying price change.

Figure 9.1 shows the relationship between the price of a standard European call option, $c(F)$, on an energy forward contract and the price of that underlying forward. The diagonal straight line at the price F_t is the slope of the option price curve and is approximately given by the ratio of the change in the option price Δc for a small change in the underlying forward price ΔF , centred on the current price. This ratio, $\Delta c/\Delta F$, is known as the *delta* of the option and measures the sensitivity of the option price to changes in the underlying (here the forward price).

The delta of the option tells us the number of units of the underlying such that the position in the underlying will change by the same amount as the option for a small change in the underlying unit price. Thus in order to offset a change in the option price we must take a position in the underlying which is equal to the negative of the option delta. A specific example will make this clear. Suppose we have written a call option. The delta of a short position in a call option is equal to $-\Delta c/\Delta F$ and therefore to delta hedge

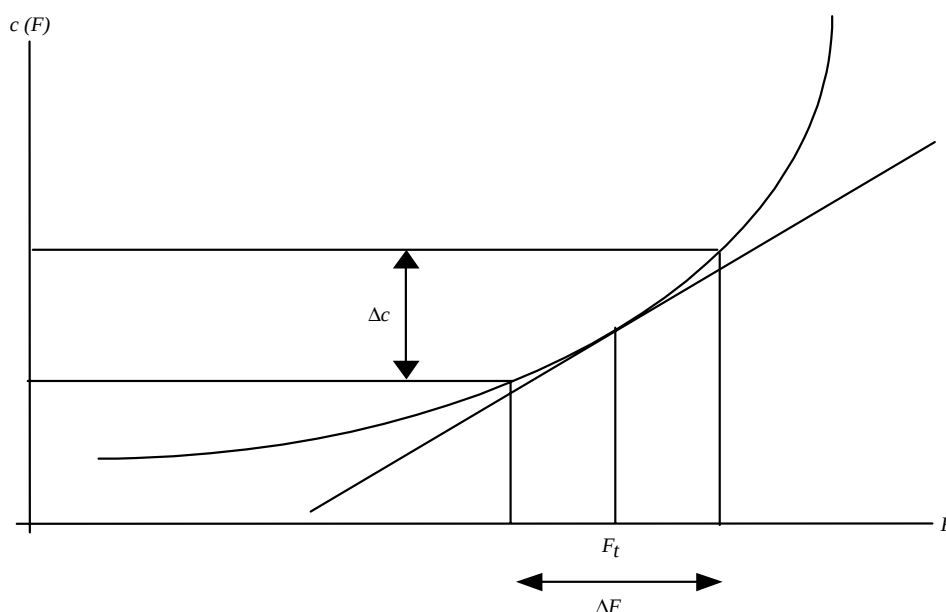


FIGURE 9.1 Sensitivity of a Call Option Price with Respect to Forward Price

this position we must buy $\Delta c/\Delta F$ of the underlying forward contract. If P denotes the value of the hedged portfolio, then we have¹;

$$P = -c + \frac{\Delta c}{\Delta F} F \quad (9.1)$$

If the forward price were to change by a small amount ΔF the change in the value of the portfolio will therefore be

$$\Delta P = -\Delta c + \frac{\Delta c}{\Delta F} \Delta F = 0 \quad (9.2)$$

The change in the portfolio value is therefore equal to zero and so the portfolio is hedged against small changes in the underlying price and the portfolio is said to be delta neutral. Theoretically, in order for the hedge to be perfect we must consider the changes in F to become very small which leads to

$$\left. \frac{\Delta C}{\Delta F} \right|_{\Delta F \rightarrow 0} = \frac{\partial C}{\partial F} = \text{delta} \quad (9.3)$$

In equation (9.3) C is now taken to be the price of any general energy derivative. Since delta changes continuously as the underlying forward price changes, this implies that we must continuously trade the underlying contract in order to keep our position in the underlying equal to delta and our portfolio hedged. In reality it is not possible or necessary to trade the underlying continuously but only when the underlying has moved by a significant amount. We discuss the issue of the frequency with which the hedge should be rebalanced later.

If the delta hedging process is continued to the option expiry date then the hedger is left with an overall position in cash and the underlying which exactly offsets the payoff of the option. If the option finishes out-of-the-money both the cash and underlying position are zero. If the option finishes in the money then the cash and underlying position is equal to the difference between the underlying price and the strike price (i.e. the payoff to the option).

Consider the simple one-factor forward price model of section 6.2. Recall that the price of a standard European call futures option, $c(t, F, K, T, s)$, with strike price K , and maturity T is given by

$$c(t, F, K, T, s) = P(t, T)[F(t, s)N(h) - KN(h - \sqrt{w})] \quad (9.4)$$

where

$$h = \frac{\ln(F(t, s)/K) + \frac{1}{2}w}{\sqrt{w}}, \quad w = \frac{\sigma^2}{2\alpha} (e^{-2\alpha(s-T)} - e^{-2\alpha(s-t)})$$

¹ Time subscripts on the forward and option prices have been dropped for ease of exposition

For this model the delta of the option is therefore given explicitly by

$$\frac{\partial C(t, F, K, T, s)}{\partial F(t, s)} = P(t, T)N(h) \quad (9.5)$$

For the same standard European call futures option but evaluated under the general multi-factor model of chapter 8 the delta is given by equation (9.5) with

$$w = \sum_{i=1}^n \left\{ \int_t^T \sigma_i(u, T)^2 du \right\} \quad (9.6)$$

Figure 9.2 plots the delta profile for two at-the-money European call forward options on NYMEX crude oil with maturities of 3 months and 6 months on an underlying forward with nine months to maturity. The forward price underlying the option is assumed to be 23.97 which was the price of the nine month forward on 31 March 2000. We use a three factor model with the volatility functions taken from the historical estimation in section 8.6.

When the forward price is low relative to the strike price, the delta of the option is close to zero reflecting the low probability of the option finishing in the money. As the forward price increases, the delta of the option increases. For at-the-money options, where the strike price is close to the forward price the delta is approximately 0.5. For high forward prices the delta is close to one as the probability of finishing in-the-money

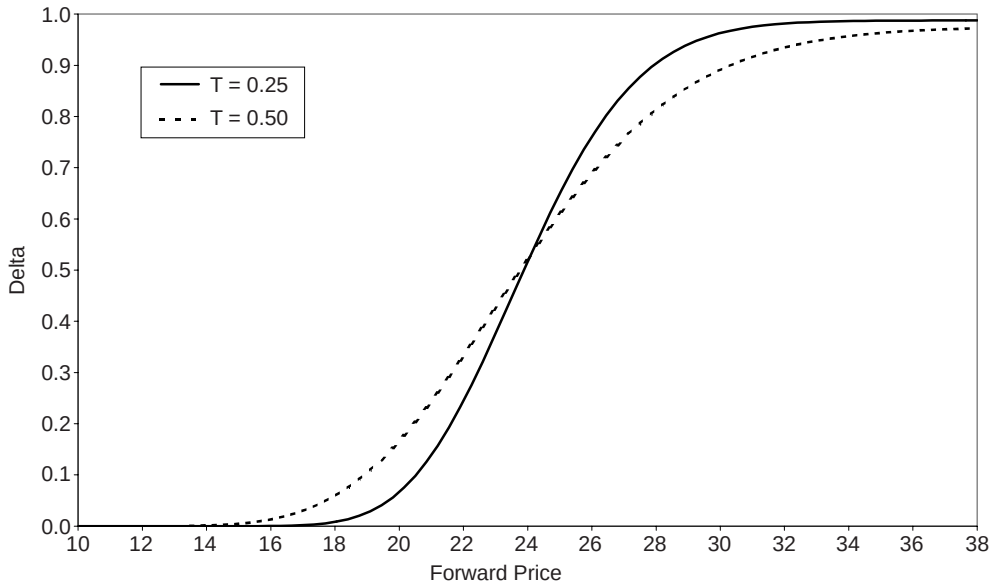


FIGURE 9.2 Delta Profile for European Call Futures Options

rate 0.05	K 24					
Volatility Functions				Date	31/03/00	03/04/00
t	σ1	σ2	σ3	T	0.50	0.4918
0.0000	0.3257	-0.1531	-0.0751	s	0.75	0.7418
0.0833	0.3076	-0.1056	0.0063	F(0,s)	23.97	23.32
0.1667	0.2885	-0.0778	0.0180	Call	-1.6210	-1.2930
0.2500	0.2729	-0.0554	0.0169	Delta	-0.5190	-0.4576
0.3333	0.2613	-0.0395	0.0189	Position in F	0.5190	0.5190
0.4167	0.2525	-0.0273	0.0231	Δcall		0.3280
0.5000	0.2448	-0.0151	0.0208	ΔF		-0.3373
0.5833	0.2367	-0.0049	0.0165	Error		-0.0093
0.6667	0.2298	0.0046	0.0147			
0.7500	0.2257	0.0115	0.0147			
0.8333	0.2188	0.0237	0.0083			
0.9167	0.2143	0.0243	0.0142			
1.0000	0.2087	0.0322	0.0078			

FIGURE 9.3 Delta Hedging a European Call Option on a NYMEX Crude Oil Futures Contract

is high. Figure 9.2 also shows that the delta becomes steeper as the option maturity decreases as the probability of the option finishing in-the-money becomes more sensitive to small changes in the forward price close to the strike price.

In figure 9.3 we show the calculations for delta hedging a short position in a six month ($T = 0.5$), at-the-money option on a 9 month NYMEX crude oil futures contract ($s = 0.75$) on the 31 March 2000 using the three factor version of the general multi-factor model. The strike price of the option is set equal to the price of the nine month futures contract (to two significant figures), $K = 24.00$. The inputs to the model are the observed forward curve on the 31 March 2000 and the three volatility functions derived from historical analysis (columns headed σ_1 , σ_2 , and σ_3).

Using equations (9.4) and (9.5) the value of the short option position is -1.6210 and the delta is -0.5190 . This delta implies that the writer of the option should take a long position of 0.5190 units of the nine month futures contract – this is indicated by the “Position in F” row in figure 9.3. Figure 9.3 also shows the performance of the delta hedge over the weekend with the hedge statistics evaluated for the closing price of the nine month futures contract on 3 April 2000. The value of the futures contract decreases to 23.32, the short option position therefore increases to -1.2930 . The option position has increased in value by 0.3280 and the position in the futures contract has decreased in value by 0.3373. Therefore the error in the hedge is the difference between changes in the values of the two positions, i.e. -0.0093 . Figure 9.4 graphs the delta hedge’s performance for a wide range of future prices. The hedge can be seen to work well close to the current

futures price (D_{call} denotes the change in the option price with DF denoting the change in the futures price) but declines in effectiveness as the forward moves away from the current forward price.

Figure 9.4 helps to answer the question of how often should the delta hedge be rebalanced. The answer is not terms of a time interval but in terms of how much the underlying price has moved from the level at which the hedge was established. Alternatively, we could set the hedge rebalancing points or triggers in terms of an upper limit on the hedging error. The hedge rebalancing trigger points are not the only factor to consider. In practice, every time the hedge is rebalanced, costs are incurred in trading in the underlying asset. These costs, for example bid-ask spreads and liquidity costs, can be significant in energy markets where even the futures markets can have relatively wide spreads and liquidity problems. Efficient hedging therefore requires an appropriate trade-off between risk reduction and trading costs. This trade-off can be analysed by simulating proposed hedging strategies using historical time series of market prices and Monte Carlo simulation.

Now consider hedging a standard European swaption, $Swpn(t;K,T,\Delta T,m)$, with strike price K that matures at time T on a swap with m resets of period ΔT . Recall from chapter 8 that there does not exist a closed-form solution for the price of a European swaption in the general multi-factor model. Therefore there will also not be a closed form solution for the delta. The natural hedge instruments are the underlying forward contracts of the swap. The delta positions are therefore given by the sensitivity of the option price with

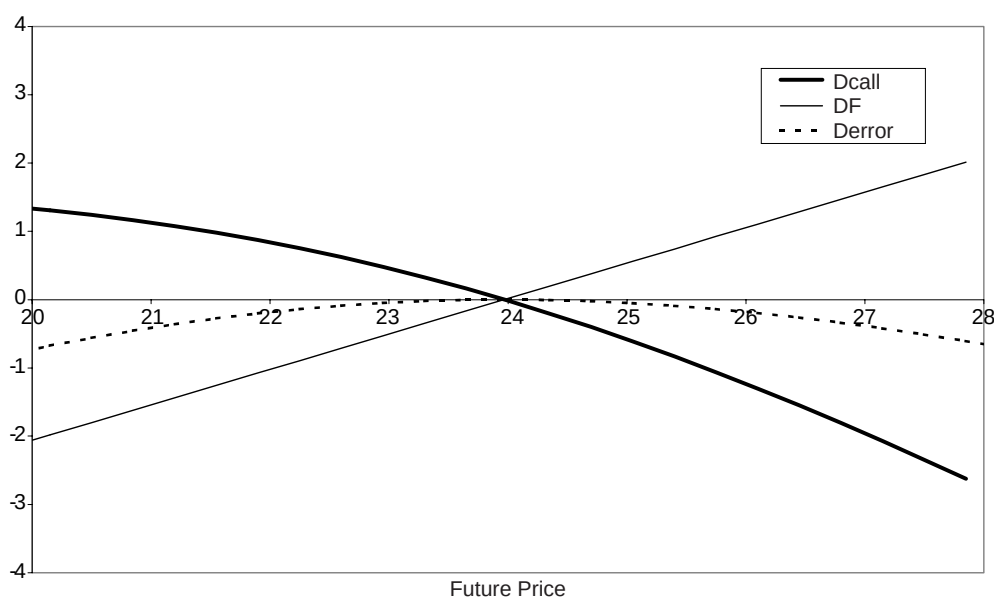


FIGURE 9.4 Performance of a Futures Option Delta Hedge Over a Weekend for a Range of Futures Prices

respect to the forward contracts which must be evaluated numerically;

$$\frac{\partial Swpn(t; K, T, \Delta T, m)}{\partial F(t, s_j)}; j = 1, \dots, m \quad (9.7)$$

For a swap with a large number of reset dates, delta hedging using all of the underlying forwards would be impractical or even impossible if they do not trade on a futures exchange. We show in section 9.5 how to deal with this problem.

9.3 Gamma Hedging

The reason for the declining effectiveness of the delta hedge as the underlying price moves away from the level at which the hedge was established can be understood from figure 9.4. The option is a non-linear (curved) function of the futures price whilst the value of the delta hedge, a fixed position in the futures contract, is a linear function. Therefore, as the underlying price moves by larger amounts the differences in the value of the option and delta hedge are greater and do not exactly offset. Another way to view this problem is that, as the underlying price changes, the delta of the option changes and so the position in the underlying required to be delta neutral changes – i.e. the delta hedge is sensitive to changes in the underlying asset. This problem is most severe for options whose strike price is close to the current underlying price. As figure 9.2 illustrates, for options which are far out-of-the-money the option price does not change very much when the underlying price changes ($\text{delta} \cong 0$) and for options which are far in-the-money the option price changes in a linear way ($\text{delta} \cong 1$).

We can solve this problem by neutralising the sensitivity of our delta hedge to changes in the underlying price. In order to do this we need to make the sensitivity of the delta of the hedged portfolio equal to zero. The sensitivity of delta to changes in the underlying asset is known as *gamma*. The calculation of gamma can be performed in a similar way to delta, as a finite difference approximation or as the second derivative of the option price with respect to the underlying price;

$$\text{gamma} = \frac{\partial^2 C}{\partial F^2} = \frac{\left(\frac{C(F+\Delta F) - C(F)}{\Delta F} \right) - \left(\frac{C(F) - C(F-\Delta F)}{\Delta F} \right)}{\Delta F} \bigg|_{\Delta F \rightarrow 0} \quad (9.8)$$

A low value of gamma implies that the option delta changes only slowly and so a delta hedge would not need to be rebalanced very frequently for it to remain delta neutral. A large gamma implies that the delta changes rapidly for changes in the underlying price, and so the hedge would have to be adjusted frequently. Figure 9.2 showed that the delta for a European call futures option remained relatively unchanged for deep in-the-money and deep out-of-the-money options, but could move quickly for at-the-money options. Figure 9.5 plots the gamma for the same two options as used in figure 9.2.

For standard European futures options, gamma can be obtained analytically in an

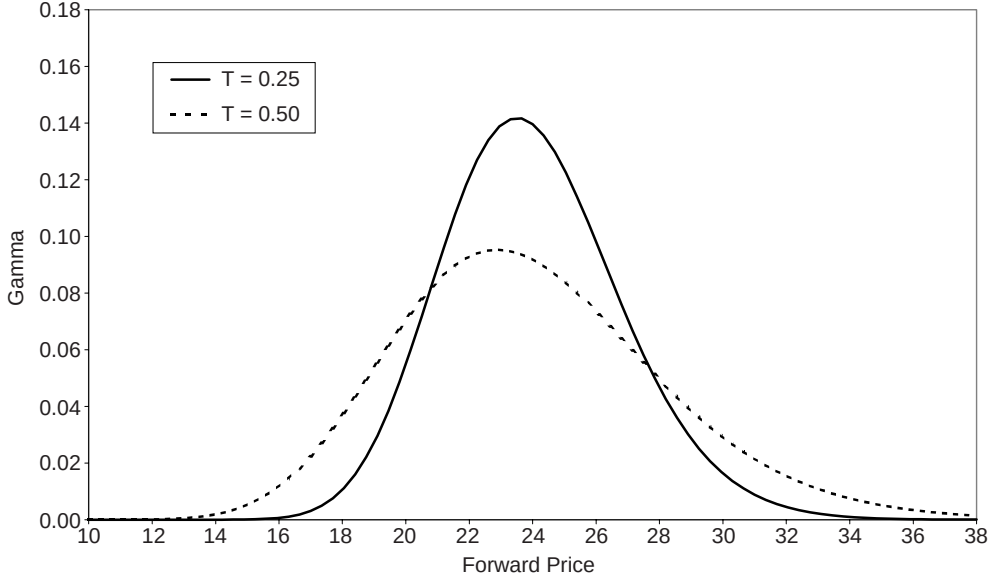


FIGURE 9.5 Gamma Profile for European Call Future Options

analogous form to the Black-Scholes version;

$$\frac{\partial^2 c(t, F, K, T, s)}{\partial F(t, s)^2} = \frac{P(t, T)n(h)}{F(t, s)\sqrt{w}} \quad (9.9)$$

where $n(\cdot)$ is the standard normal density function and w is given as in equation (9.4) for the single factor model of section 8.2 and as in equation (9.6) for the multi-factor model of section 8.4.

In order to neutralise the gamma of a portfolio we must use another option since the gamma of a forward or futures contract is zero. Let C_{target} represent the value of the option we are trying to hedge (the ‘target’ option) and Δ_{target} and Γ_{target} the delta and gamma of the target option respectively. Now, let C_{hedge} , Δ_{hedge} and Γ_{hedge} represent the price, delta, and gamma of the option which we use to hedge the target option (the ‘hedge’ option). In order for our option position to be simultaneously delta-gamma neutral we first choose a position β in the hedge option such that the gamma of the portfolio is zero, i.e. we require;

$$\Gamma_{target} + \beta\Gamma_{hedge} = 0 \quad (9.10)$$

Equation (9.10) implies that the position that has to be taken in the hedge option to make the portfolio delta-gamma neutral is $\beta = -\Gamma_{target}/\Gamma_{hedge}$. The combined position in the target and hedge options, by construction, has a gamma of zero but a non-zero

<table><tr><td>rate</td></tr><tr><td>0.05</td></tr></table>				rate	0.05	<table><tr><td>K</td></tr><tr><td>24</td></tr></table>				K	24
rate											
0.05											
K											
24											
Volatility Functions				Fri		Mon					
				Date	31/03/00		03/04/00				
t	σ_1	σ_2	σ_3		Target	Hedge	Target	Hedge			
0.0000	0.3257	-0.1531	-0.0751	T	0.50	0.25	0.4918	0.2418			
0.0833	0.3076	-0.1056	0.0063	s	0.75	0.75	0.7418	0.7418			
0.1667	0.2885		0.0180								
0.2500	0.2729	-0.0554	0.0169	F(0,s)	23.97	23.97	23.32	23.32			
0.3333	0.2613	-0.0395	0.0189								
0.4167	0.2525	-0.0273	0.0231	Call	-1.6210	1.0936	-1.2930	0.7740			
0.5000	0.2448	-0.0151	0.0208	Delta	-0.5190	0.5127	-0.4576	0.4192			
0.5833	0.2367	-0.0049	0.0165	Gamma	-0.0922	0.1399	-0.0955	0.1436			
0.6667	0.2298	0.0046	0.0147								
0.7500	0.2257	0.0115	0.0147	β	0.6591		0.6591				
0.8333	0.2188	0.0237	0.0083	Residual Delta	-0.1811						
0.9167	0.2143	0.0243	0.0142	Position in F	0.1811		0.1811				
1.0000	0.2087	0.0322	0.0078								
				Δ call(Target)			0.3280				
				Δ call(Hedge)			-0.2106				
				ΔF			-0.1177				
				Error			-0.0003				

FIGURE 9.6 Delta-Gamma Hedge for European Call Option on a NYMEX Crude Oil Futures Contract

residual delta since, in general, the delta's of the target and hedge option positions will not exactly cancel. We can neutralise the delta of the combined portfolio of the two options by taking a position in the underlying asset equal to the negative of the residual delta. This delta hedge position will not affect the portfolio's gamma because the underlying asset has a gamma of zero. Note that the gamma must be eliminated first and then the residual delta. Figure 9.6 shows the calculations for setting up a delta-gamma hedge for the same target option as in figure 9.3

The short position in the target option has a value of -1.6210 , a delta of -0.5190 and a gamma of -0.0922 . For the hedge option we use a three month maturity European call with otherwise the same characteristics as the target option. This option has a price of 1.0936 , a delta of 0.5127 , and a gamma of 0.1399 . In order to neutralise the gamma of the portfolio the user buys $-(-0.0922)/(0.1399) = 0.6591$ of the hedge option. The residual delta of the portfolio is $(-0.5190) + 0.6591 \times (0.5127) = -0.1811$. Therefore a long position of 0.1811 futures contracts must be taken to eliminate the delta of the portfolio. Figure 9.6 also shows the performance of the hedge over the weekend. From the 31 March 2000 to the 3 April 2000 the futures price decreases from 23.97 to 23.32 causing the short position in the target option to increase in value by 0.3280 . The change in the long hedge position decreases in value by 0.2106 and the long position in the future

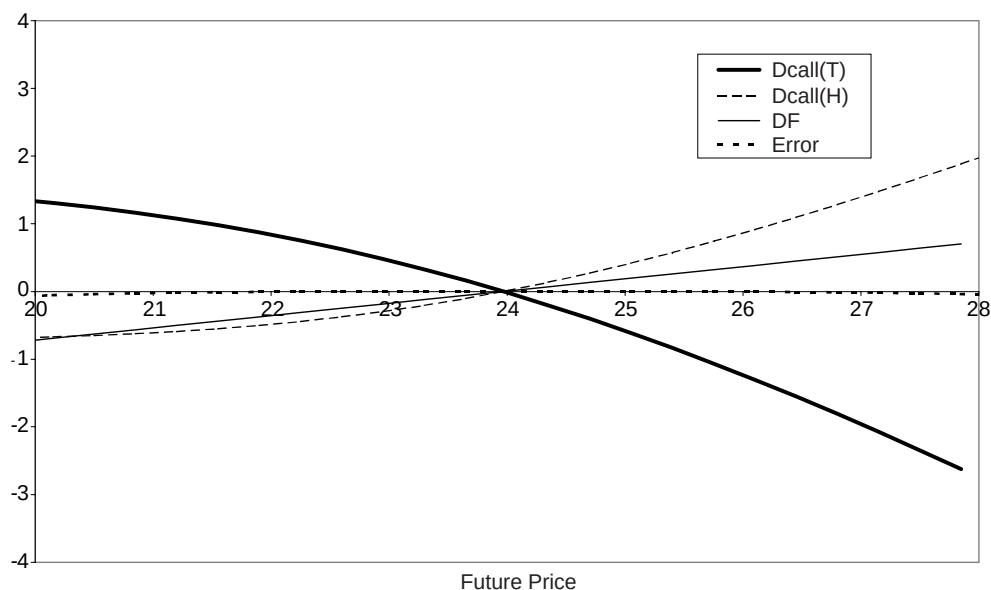


FIGURE 9.7 Performance of a Futures Option Delta-Gamma Hedge Over a Weekend for a Range of Futures Prices

decreases by 0.1177, leading to a hedging error of -0.0003 to the 4th decimal place. Figure 9.7 shows the performance of the delta-gamma hedge for a range of values of the underlying futures contract. By comparison with figure 9.4, the delta-gamma hedging error is significantly smaller than the delta hedging error for a wide range of futures prices.

Since the hedging error is small for a wide range of futures prices the hedge needs to be rebalanced much less frequently. However, trading costs in options markets are typically much greater than in the futures markets, therefore it is still important to compare the improvement in the hedge gained by gamma hedging with the additional cost involved in trading in the options market.

9.4 Hedging Volatility and Other Sources of Market Risk

In addition to the risk due to changes in the spot or forward prices, the other major source of market risk comes from changes in volatility. We have seen throughout the previous chapters that, apart from price, the volatility or uncertainty in the price of the underlying energy is the key determinant of the value of derivatives. Thus, changes in volatility are an important source of risk in a derivatives portfolio. Volatility can be hedged using a similar technique to the gamma hedging described in the previous section. The sensitivity of an option or portfolio to changes in volatility is called *vega* and

can be calculated in a similar way to delta;

$$vega = \frac{\partial C}{\partial \sigma_F} = \left(\frac{C(\sigma_F + \Delta\sigma_F) - C(\sigma_F - \Delta\sigma_F)}{2\Delta\sigma_F} \right) \Big|_{\Delta\sigma_F \rightarrow 0} \quad (9.11)$$

In order to neutralise vega we must take a position in an additional hedging option such that the vega of the hedging option exactly cancels the vega of the original target portfolio. The steps involved in constructing a delta-vega hedge are identical to those in the previous section for a delta-gamma hedge but with gamma replaced by vega. However, in many cases a trader may want to neutralise delta, gamma and vega. This requires trading in two different hedging options, say ‘hedge1’ and ‘hedge2’. Since options have both gamma and vega, we must neutralise both gamma and vega simultaneously by solving the following two equations;

$$\Gamma_{\text{target}} + \beta_{\text{hedge1}} \Gamma_{\text{hedge1}} + \beta_{\text{hedge2}} \Gamma_{\text{hedge2}} = 0 \quad (9.12)$$

$$V_{\text{target}} + \beta_{\text{hedge1}} V_{\text{hedge1}} + \beta_{\text{hedge2}} V_{\text{hedge2}} = 0 \quad (9.13)$$

The solution to equations (9.12) and (9.13) is;

$$\beta_{\text{hedge1}} = \frac{\Gamma_{\text{hedge2}} V_{\text{target}} - \Gamma_{\text{target}} V_{\text{hedge2}}}{\Gamma_{\text{hedge1}} V_{\text{hedge2}} - \Gamma_{\text{hedge2}} V_{\text{hedge1}}} \quad (9.14)$$

$$\beta_{\text{hedge2}} = \frac{\Gamma_{\text{target}} V_{\text{hedge1}} - \Gamma_{\text{hedge1}} V_{\text{target}}}{\Gamma_{\text{hedge1}} V_{\text{hedge2}} - \Gamma_{\text{hedge2}} V_{\text{hedge1}}} \quad (9.15)$$

With these solutions for the positions in the hedging options the residual delta can be calculated to obtain the position required in the underlying energy asset.

9.5 Factor Hedging

In this section we describe a general approach to hedging a portfolio of energy derivatives. The method is based on the multi-factor model described in chapter 8. The set of factors, either estimated from historical data or implied from the market prices of options, represent the risks to which a portfolio of energy derivatives is exposed. For example, figure 9.8 (which reproduces figure 8.8) shows the three main factors (volatility functions) for the NYMEX crude oil forward curve.

A portfolio of energy derivatives; for example caps and swaptions, is exposed to the risk that the crude oil forward curve may change by a combination of random proportions of these “shifting”, “tilting” and “bending” factors. In order to hedge the energy portfolio we need to take positions in a set of hedge instruments (typically futures and futures options) such that the hedged portfolio is immunised against the possible changes in the forward curve described by the volatility functions.

The first step in the process is to work out how the portfolio changes in value if the forward curve were to be shocked by each of the volatility functions separately. The

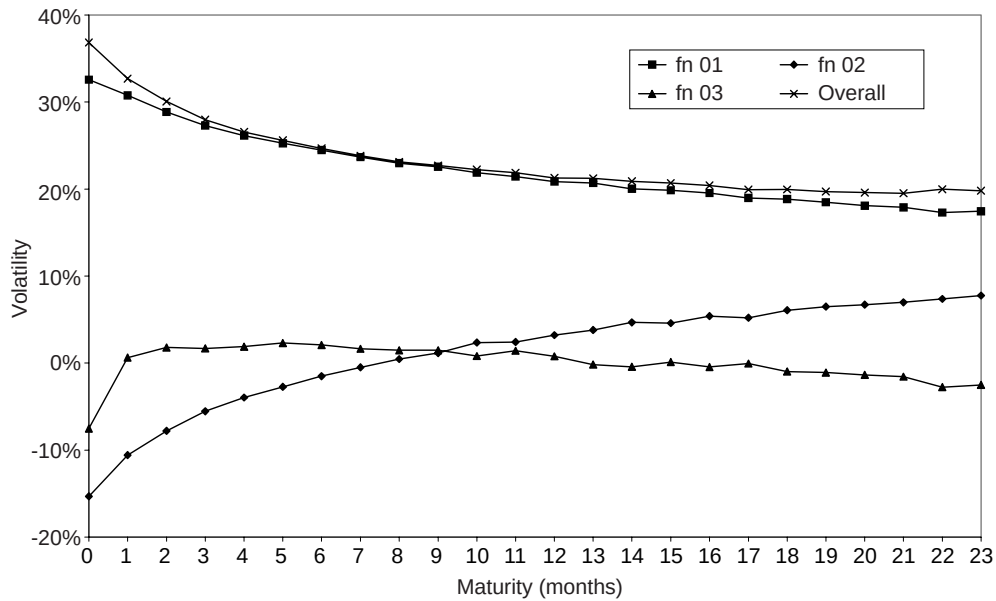


FIGURE 9.8 Three Main Volatility Functions for NYMEX Crude Oil Forward Curve

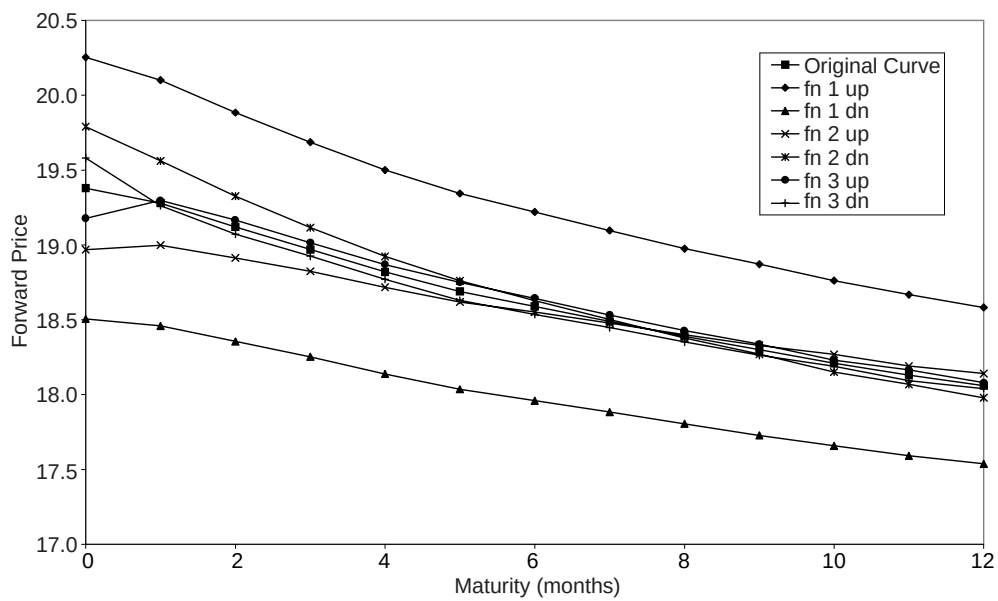


FIGURE 9.9 Shifting the NYMEX Crude Oil Forward Curve by its Three Main Factors

appropriate shocks to apply to the forward curve are obtained by discretising the multi-factor model of equation (8.4) taking each volatility function in turn:

$$\Delta F_i(0, s) = F_i(0, s) \sigma_i(0, s) \Delta z_i ; \quad i = 1, \dots, 3 \quad (9.16)$$

The size of the shock Δz_i should be chosen to give a typical movement of the curve over the hedging period, Δt , for example a one standard deviation move i.e. $\sqrt{\Delta t}$ is reasonable. We discuss the choice of the hedging interval Δt and the shock Δz_i later, but for now we take a hedging interval of one week to give relatively large movements in the curve. The curves of changes described by equations (9.16) are applied in both positive and negative amounts to the current forward curve, $F(0, s)$. Figure 9.9 illustrates the result of this for the NYMEX crude oil futures curve on the 31 March 2000. In figure 9.9 'fn1 up' and 'fn1 dn' denote the original forward curve after an up shift determined by fn1 and a down shift respectively. The other shifts are defined similarly.

The next step in the process is to compute the changes in the value of the portfolio between the downward and upward shifts of the forward curve for each factor. If P represents the value of the portfolio, we have;

$$\Delta P_i = P(F_{i,U}(\cdot)) - P(F_{i,D}(\cdot)) ; \quad i = 1, \dots, 3 \quad (9.17)$$

In equation (9.17) the U subscript indicates the up shifted forward curve and D indicates the down shifted curve. The calculation of the changes in the portfolio value require complete revaluation of the portfolio based on the shifted forward curves.

The three changes in the portfolio can be hedged using three different forward contracts $F(0, s_i)$ choosing appropriate positions β_i in these contracts such that the overall change in the hedged portfolio P_H is zero for each factor. That is we solve the following set of equations for the unknown positions β_i ;

$$\begin{aligned} \Delta P_{H,1} &= \Delta P_1 + \beta_1 \Delta F(0, s_1)_1 + \beta_2 \Delta F(0, s_2)_1 + \beta_3 \Delta F(0, s_3)_1 = 0 \\ \Delta P_{H,2} &= \Delta P_2 + \beta_1 \Delta F(0, s_1)_2 + \beta_2 \Delta F(0, s_2)_2 + \beta_3 \Delta F(0, s_3)_2 = 0 \\ \Delta P_{H,3} &= \Delta P_3 + \beta_1 \Delta F(0, s_1)_3 + \beta_2 \Delta F(0, s_2)_3 + \beta_3 \Delta F(0, s_3)_3 = 0 \end{aligned} \quad (9.18)$$

Equations (9.18) are a set of three linear equations in the three unknowns, $\beta_i ; i = 1, \dots, 3$, which can be solved easily and exactly as a matrix equation. An alternative and more general solution method is simply to minimise the sum of the squared hedging errors (ΔP_H);

$$\text{minimise } \left\{ (\Delta P_{H,1})^2 + (\Delta P_{H,2})^2 + (\Delta P_{H,3})^2 ; \beta_1, \beta_2, \beta_3 \right\} \quad (9.19)$$

A specific example will help to clarify the method. We assume the portfolio to be hedged contains a single European one month, at-the-money crude oil swaption on a six month swap with monthly resets. We hedge the three crude oil factors identified in section 8.6 using the nearest, fourth nearest and seventh nearest futures contracts. Figure 9.10

The hedge valuation date is 31 = 1/12) and exercises into a six ng payments are swapped for a a, assuming interest rates of 5%

'Shifted Curves' contain the information presented by equation (9.16). The volatility (σ), P , and the price of the option (P_0), and the changes in the portfolio (ΔP),

curves	Down Shifted Curves		
fn3 up	fn1 dn	fn2 dn	fn3 dn
19.1783	18.5059	19.7910	19.5817
19.2967	18.4587	19.5621	19.2633
19.1677	18.3560	19.3261	19.0723
19.0143	18.2531	19.1157	18.9257
18.8692	18.1389	18.9230	18.7708
18.7497	18.0364	18.7607	18.6303
18.6436	17.9597	18.6288	18.5364
18.5323	17.8839	18.5024	18.4477
18.4274	17.8047	18.3782	18.3526
18.3374	17.7281	18.2709	18.2626
18.2308	17.6583	18.1502	18.1892
18.1656	17.5920	18.0690	18.0944
18.0796	17.5381	17.9795	18.0404

0.6074	0.2967	0.6324	0.5566
19.1783	18.5059	19.7910	19.5817
19.0143	18.2531	19.1157	18.9257
18.6436	17.9597	18.6288	18.5364

fn3
0.0508
-0.4034
0.0886
0.1071

0.0000

-0.2248

Risk Management of Energy Derivatives

Date		31/03/00		Az		0.1385												
				Volatility Functions					Up Shifted Curves					Down Shifted Curves				
		s	F(0,s)	fn1	fn2	fn3	fn1 up	fn2 up	fn3 up	fn1 dn	fn2 dn	fn3 dn						
s1		0.0000	19.38	0.3257	-0.1531	-0.0751	20.2541	18.9690	19.1783	18.5059	19.7910	19.5817						
	T1	0.0833	19.28	0.3076	-0.1056	0.0063	20.1013	18.9979	19.2967	18.4587	19.5621	19.2633						
T2		0.1667	19.12	0.2885	-0.0778	0.0180	19.8840	18.9139	19.1677	18.3560	19.3261	19.0723						
s2		0.2500	18.97	0.2729	-0.0554	0.0169	19.6869	18.8243	19.0143	18.2531	19.1157	18.9257						
T3		0.3333	18.82	0.2613	-0.0395	0.0189	19.5011	18.7170	18.8692	18.1389	18.9230	18.7708						
s3		0.4167	18.69	0.2525	-0.0273	0.0231	19.3436	18.6193	18.7497	18.0364	18.7607	18.6303						
		0.5000	18.59	0.2448	-0.0151	0.0208	19.2203	18.5512	18.6436	17.9597	18.6288	18.5364						
		0.5833	18.49	0.2367	-0.0049	0.0165	19.0961	18.4776	18.5323	17.8839	18.5024	18.4477						
		0.6667	18.39	0.2298	0.0046	0.0147	18.9753	18.4018	18.4274	17.8047	18.3782	18.3526						
		0.7500	18.3	0.2257	0.0115	0.0147	18.8719	18.3291	18.3374	17.7281	18.2709	18.2626						
		0.8333	18.21	0.2188	0.0237	0.0083	18.7617	18.2698	18.2308	17.6583	18.1502	18.1892						
		0.9167	18.13	0.2143	0.0243	0.0142	18.6680	18.1910	18.1656	17.5920	18.0690	18.0944						
	1.0000	18.06	0.2087	0.0322	0.0078	18.5819	18.1405	18.0796	17.5381	17.9795	18.0404							

Portfolio		K	T	P	0.9881	0.5334	0.6074	0.2967	0.6324	0.5566		
Swaption				F(0,s1)	20.2541	18.9690	19.1783	18.5059	19.7910	19.5817		
K	18.78			F(0,s2)	19.6869	18.8243	19.0143	18.2531	19.1157	18.9257		
T	0.0833			F(0,s3)	19.2203	18.5512	18.6436	17.9597	18.6288	18.5364		
dT	0.0833			c(K1,T1)	1.1435	0.5831	0.7094	0.3579	0.7957	0.6602		
nr	6	19.12	0.0833	c(K2,T2)	1.3368	0.8475	0.9447	0.5926	0.9990	0.8986		
rate	0.05	18.97	0.1667	c(K3,T3)	1.5703	1.1645	1.2327	0.8865	1.2386	1.1702		
		18.69	0.3333									

P0		0.5817		fn1 up	fn2 up	fn3 up	fn1 dn	fn2 dn	fn3 dn
c(K1,T1)	0.6845	ΔP		0.4065	-0.0483	0.0257	-0.2850	0.0508	-0.0251
c(K2,T2)	0.9215	ΔF(s1)		0.8741	-0.4110	-0.2017	-0.8741	0.4110	0.2017
c(K3,T3)	1.2012	ΔF(s2)		0.7169	-0.1457	0.0443	-0.7169	0.1457	-0.0443
		ΔF(s3)		0.6303	-0.0388	0.0536	-0.6303	0.0388	-0.0536
		Δc(1)		0.4590	-0.1014	0.0249	-0.3266	0.1112	-0.0243
		Δc(2)		0.4153	-0.0740	0.0232	-0.3289	0.0776	-0.0229
		Δc(3)		0.3691	-0.0367	0.0315	-0.3148	0.0373	-0.0311
		ΔPH		0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
		beta		-0.0199	0.3250	0.1831	0.1584	-0.9190	-1.1609

FIGURE 9.11 Delta-Gamma Factor Hedging a European Swaption

the hedge instruments ($\Delta F(.)$) and the hedged portfolio (ΔPH) are shown towards the bottom centre of figure 9.10. In the box labelled 'beta' is the solution for the positions to be held in the hedge futures such that the changes in the hedged portfolio are zero for all three factor shifts.

The approach we have just described can be seen as a general form of delta hedging. It

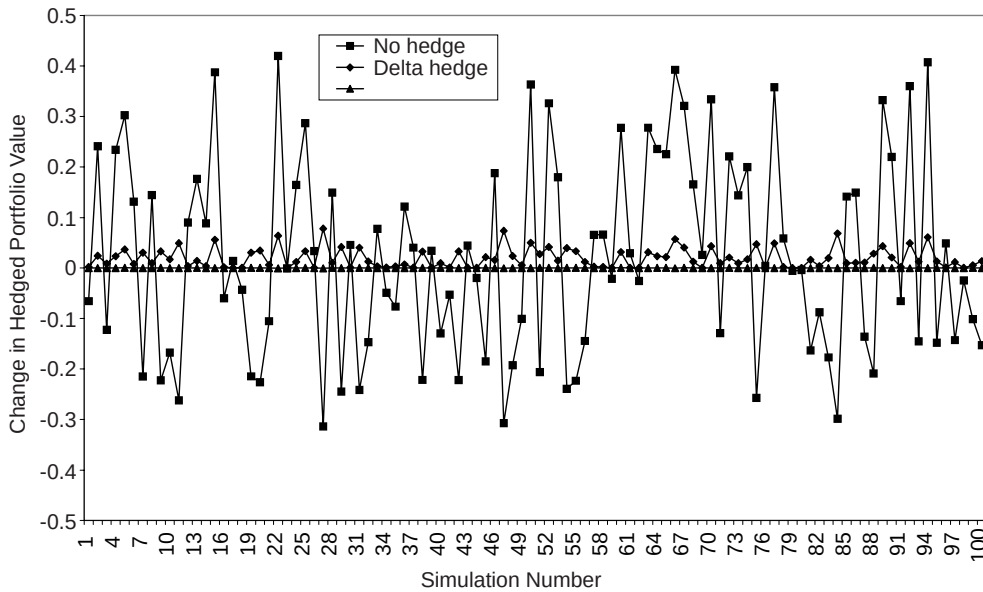


FIGURE 9.12 Performance of Factors Hedges

allows us to delta hedge the factors driving the entire forward curve. The method suffers from the same problem as the simple delta hedge discussed in section 9.2. Since we are hedging a portfolio of derivatives, which change in a non-linear way with respect to the forward curve, with instruments that have linear payoffs, the hedge is only good for small changes in the forward curve. We can assess the hedging error in a similar way to that used in sections 9.2 and 9.3, by varying the size of the changes in the forward curve via Δz . Using this approach we can determine how far we can allow the forward curve to move, in terms of Δz , before the factor hedge must be rebalanced. This also indicates the size of Δz which should be used in calculating the factor hedge.

The factor delta hedge can be improved in a similar way as for the simple delta hedge – by using standard European futures options to gamma hedge the factors. Continuing with our previous example we now introduce as hedge options three European futures at-the-money call options with maturities T_1 , T_2 and T_3 and strike prices K_1 , K_2 and K_3 . Figure 9.11 shows the calculations for the delta-gamma factor hedge.

We use the same six shifted forward curves but we now consider the shifts relative to the current forward curve. This gives six changes in the portfolio and the six hedge instruments (three futures and three futures options). We compute the six positions, β_i ; $i = 1, \dots, 6$, in these hedge instruments so as to minimise the sum of the squared changes in the hedged portfolio. To demonstrate the performance of both the delta and delta-gamma factor hedges, figure 9.12 shows the changes in the value of the hedged portfolio for 100 simulated changes in the forward curve under the combined three factors for the cases of no hedge, the delta factor hedge and the delta-gamma factor hedge.

Over these 100 simulations the unhedged portfolio changes range between -31 cents and $+42$ cents, narrowing to -0.01 cents and $+7$ cents for the delta hedged portfolio and -0.08 cents and $+0.05$ cents for the gamma hedged portfolio

9.6 Summary

In this chapter we discussed the risk management of portfolios that contain energy derivatives. Firstly, we introduced the basic concepts of delta, gamma and vega hedging for a single option position. We then described how the multi-factor forward curve model developed in chapter 8 could be used to generalise the delta and gamma hedging methods to handle a complex portfolio of energy derivatives. This factor hedging approach was illustrated with the example of hedging a crude oil swaption using futures and futures options. Finally, we illustrated how the effectiveness of the factor hedge could be assessed via Monte Carlo simulation.